Presentation Script & Transcript

Impact of Data Augmentation on CNN Performance and Robustness

English + Roman Urdu, slide by slide

Student	CMS	Slides (suggested)
Sohaib Aamir	538108	1 – 12
Abdul Rafeh	450126	13 – 23
Mahmoob Ul Hassan	624084	24 – 35

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How to use this document

- **Part A**: what to say on each of the 35 slides. Every slide has an **English** script (to read in class) and the same content in **Roman Urdu** (to understand and remember it). About 40–60 seconds per slide; ~25–30 minutes in total.
- **Part B**: Sir's Lecture 7 transcript, where the topic was assigned: original in Roman Urdu, with a full English translation.
- **Part C**: the 10 questions Sir is most likely to ask, with short answers in both languages.
- The speaker split on the cover is only a suggestion: theory → method → results. Change it as you like.

Part A · Slide-by-slide script

Slide 1 · Title — Sohaib

ENGLISH

Assalam-o-Alaikum. We are Sohaib Aamir, Abdul Rafeh and Mahmoob Ul Hassan. Our topic is the impact of data augmentation on CNN performance and robustness. In Lecture 7 we warped images with 3×3 matrices. We show that the same matrices, applied at random during training, make a CNN more accurate and much more robust, and we measured exactly by how much.

ROMAN URDU

Assalam-o-Alaikum. Hum Sohaib Aamir, Abdul Rafeh aur Mahmoob Ul Hassan hain. Hamara topic hai: data augmentation ka CNN ki performance aur robustness par asar. Lecture 7 mein humne images ko 3×3 matrices se warp kiya tha. Hum dikhayenge ke wahi matrices agar training mein random lagayi jayen to CNN zyada accurate aur kahin zyada robust ho jata hai, aur humne exactly naapa hai ke kitna.

Slide 2 · Introduction — Sohaib

ENGLISH

This topic came from Lecture 7, on the similarity and affine slides, where Sir said: “Toh jo image hai na, uski na ye saari augmentation hai.” Every transformation we wrote as a 3×3 matrix is also a way to create a new, correctly-labelled training image: a cat rotated by 10 degrees is still a cat. Data augmentation means creating new training images by transforming existing ones while keeping the label. There are two kinds. Geometric augmentation changes position, size and viewpoint, so it needs warping and interpolation. Photometric augmentation changes brightness, contrast, saturation and noise, pixel by pixel, with no warping. Our task was to explain the maths, implement both on a CNN, and prove the gain with numbers.

ROMAN URDU

Ye topic Lecture 7 se aaya, similarity aur affine wali slides par, jab Sir ne kaha: “Toh jo image hai na, uski na ye saari augmentation hai.” Har transformation jo humne 3×3 matrix se likhi, wo ek nayi training image banane ka tareeqa bhi hai jiska label sahi rehta hai: 10 degree ghoomi hui billi phir bhi billi hai. Data augmentation ka matlab: maujooda images ko transform kar ke nayi images banana, label same. Do qisam: geometric (position, size, viewpoint) jis mein warping aur interpolation chahiye; aur photometric (brightness, contrast, saturation, noise) jo har pixel par alag lagti hai, warping nahi. Hamara kaam: maths samjhana, dono ko CNN par lagana, aur numbers se saabit karna.

Slide 3 · Problem statement

Sohaib

ENGLISH

Two problems motivate this work. First, overfitting with limited data: our CNN has 6.5 million parameters and can memorise 10,000 images, reaching 100% training accuracy, but only 78.5% on images it has not seen, a 21.5-point gap. Second, fragile inference: real test images are rotated, shifted, darker or noisier. Without augmentation, accuracy drops by 51% on rotated and dark images. Our goal is to use the Lecture 7 transformations as augmentation and measure, on a fixed held-out test set, how much accuracy and robustness improve. Both numbers come from our own baseline run.

ROMAN URDU

Do masle hain. Pehla, kam data par overfitting: hamare CNN mein 6.5 million parameters hain, wo 10,000 images ka ratta laga leta hai, training accuracy 100%, lekin nayi images par sirf 78.5%: 21.5 points ka gap. Doosra, kamzor inference: asal test images ghoomi hui, khiski hui, andheri ya noisy hoti hain. Augmentation ke baghair ghoomi aur andheri images par accuracy 51% gir jati hai. Hamara maqsad: Lecture 7 ki transformations ko augmentation bana kar fixed test set par naapna ke accuracy aur robustness kitni barhti hai. Ye dono numbers hamare apne baseline run se hain.

Slide 4 · Section: Theoretical foundation

Sohaib

ENGLISH

First, the theory: the transformation hierarchy from translation to homography.

ROMAN URDU

Pehle theory: translation se homography tak transformations ki hierarchy.

Slide 5 · Transformation hierarchy

Sohaib

ENGLISH

Each step adds freedom, and each step gives a new augmentation. Translation has 2 degrees of freedom and gives random shifts. Euclidean, or rigid, adds rotation: 3 DoF. Similarity adds uniform scale, so zoom: 4 DoF. Affine has six free parameters a to f, adding shear, aspect change and flip. Homography has 8 DoF and simulates a change of viewpoint. This also answers the question from class about how many points we need. Each point pair gives two equations, so the minimum is the DoF divided by two, rounded up. Rigid has 3 DoF, and one pair gives only 2 equations, so we need 2 pairs. A homography has 9 entries but is defined up to scale, so we fix the last one to 1 and get 8 unknowns: 4 pairs.

ROMAN URDU

Har qadam par azadi barhti hai aur ek nayi augmentation milti hai. Translation: 2 DoF, random shift. Euclidean (rigid): rotation add, 3 DoF. Similarity: uniform scale yani zoom, 4 DoF. Affine: chhe parameters a se f, shear, aspect aur flip. Homography: 8 DoF, viewpoint change. Class wale sawal ka jawab bhi yehi hai ke kitne points chahiye: har point pair 2 equations deta hai, to minimum = DoF ÷ 2, upar round. Rigid ke 3 DoF hain, ek pair sirf 2 equations deta hai, isliye 2 pairs. Homography ke 9 entries hain lekin wo scale tak defined hai, last entry 1 fix: 8 unknowns, 4 pairs.

Slide 6 · Transformations in action

Sohaib

ENGLISH

These images come from our own NumPy code, written from scratch. Rigid and similarity transforms keep angles; only position, orientation and size change. Affine keeps parallel lines parallel, so a square becomes a parallelogram. Homography keeps only straight lines, so a square can become any quadrilateral. Every panel uses inverse warping with bilinear interpolation. The black regions are pixels whose source falls outside the image.

ROMAN URDU

Ye images hamare apne NumPy code se hain, scratch se likha. Rigid aur similarity angles ko same rakhte hain, sirf position, rukh aur size badalta hai. Affine mein parallel lines parallel rehti hain, square parallelogram ban jata hai. Homography mein sirf seedhi lines seedhi rehti hain, square koi bhi chaar-kona shape ban sakta hai. Har panel inverse warping aur bilinear interpolation se bana. Black hissay woh pixels hain jinka source image ke bahar hai.

Slide 7 · Solving from point pairs

Sohaib

ENGLISH

Three point pairs fix an affine transform and four fix a homography. For affine, we stack two rows per pair into a 6×6 system and solve it by least squares; the error on the targets is 2×10^{-13} pixels. For homography we use the DLT: an 8×9 system $A\cdot h = 0$. The solution is the singular vector of A with the smallest singular value, and we divide by the last entry. The error is 3×10^{-10} pixels. With the minimum number of pairs the fit is exact; the error is only floating-point rounding. With more pairs, least squares averages the noise, and that leads to RANSAC.

ROMAN URDU

Teen point pairs affine ko fix karte hain, chaar homography ko. Affine: har pair se 2 rows, 6×6 system, least squares se solve; error 2×10^{-13} pixel. Homography: DLT, 8×9 system $A\cdot h = 0$; jawab A ka woh singular vector jiski singular value sab se chhoti hai, phir last entry se divide; error 3×10^{-10} pixel. Minimum pairs par fit exact hai, error sirf rounding ka. Zyada pairs hon to least squares noise average karta hai, aur aage RANSAC aata hai.

Slide 8 · Homography

Sohaib

ENGLISH

A homography simulates a new camera viewpoint. In homogeneous coordinates, x′ equals (ax + by + c) divided by (gx + hy + 1). The terms g and h make the divisor change across the image, so parallel lines can meet: that is perspective. Two images taken from the same centre of projection are related by a homography; this is where the lecture was heading. As augmentation, a small random H shows the same object from a slightly different angle. We keep g and h within ±0.0008 per pixel on 32×32 images. Homographies are also used for panoramas and in SuperPoint's homographic adaptation.

ROMAN URDU

Homography naya camera viewpoint simulate karti hai. x′ = (ax + by + c) ÷ (gx + hy + 1). g aur h ki wajah se denominator image mein jagah jagah badalta hai, to parallel lines mil sakti hain: yehi perspective hai. Same center of projection se li gayi do images ka rishta homography hai; lecture yahin ja raha tha. Augmentation mein chhoti random H = wahi object thore alag angle se. Hum 32×32 images par g, h ko ±0.0008 per pixel rakhte hain. Homography panorama aur SuperPoint ki homographic adaptation mein bhi use hoti hai.

Slide 9 · Section: Mathematical mechanics

Sohaib

ENGLISH

Now the mechanics: how a matrix actually produces a new image, through forward or inverse warping and interpolation.

ROMAN URDU

Ab mechanics: matrix se asal nayi image kaise banti hai: forward ya inverse warping, aur interpolation.

Slide 10 · Forward vs inverse warping

Sohaib

ENGLISH

In forward warping we push each source pixel to its target, $x' = H \cdot x$. With a zoom of 1.35 and a 20-degree rotation, 45% of the target pixels receive nothing: holes. Zooming in creates more target pixels than source pixels, so pushing can never fill them all. In inverse warping we pull: every target pixel asks where it came from, $x = H^{-1} \cdot x'$, and interpolates there. Each target is visited exactly once, so there are no holes. Every augmentation library uses inverse warping, and so do we.

ROMAN URDU

Forward warping mein har source pixel ko target ki taraf "push" karte hain: $x' = H \cdot x$. 1.35 zoom aur 20 degree rotation par 45% target pixels ko kuch nahi milta: holes. Zoom-in mein target pixels source se zyada hote hain, push se sab nahi bharte. Inverse warping mein "pull" karte hain: har target pixel poochta hai ke wo kahan se aaya, $x = H^{-1} \cdot x'$, aur wahan interpolate karta hai. Har target ek baar visit hota hai, koi hole nahi. Har augmentation library yehi use karti hai, aur hum bhi.

Slide 11 · Interpolation methods

Sohaib

ENGLISH

Inverse warping lands between pixels, so we interpolate. Nearest neighbour copies the closest pixel. It is fastest, but edges turn blocky. Bilinear uses 4 pixels: linear in x, then in y. This is the Q11, Q12, Q21, Q22 derivation from the lecture. It is smooth and slightly soft, and it is the default in most libraries. Bicubic uses 16 pixels: a cubic through 4 points in each of 4 rows, then one more cubic across. We use the Keys kernel with $a = -0.5$. It gives the sharpest edges but can overshoot. All three are written from scratch in geometry.py.

ROMAN URDU

Inverse warping do pixels ke beech girta hai, isliye interpolation. Nearest neighbour sab se qareeb pixel copy karta hai: sab se tez, lekin edges blocky. Bilinear 4 pixels: pehle x mein linear, phir y mein; lecture ka Q11, Q12, Q21, Q22 derivation. Smooth, thora soft, zyada tar libraries ka default. Bicubic 16 pixels: 4 rows mein har ek par cubic, phir ek aur cubic across; Keys kernel $a = -0.5$. Sab se sharp lekin overshoot ho sakta hai. Teeno geometry.py mein scratch se likhe.

Slide 12 · Interpolation results

Sohaib

ENGLISH

Here is the same warp with the three interpolators, on a 24×24 crop zoomed about 10 times and rotated 10 degrees. Nearest is blocky, bilinear is smooth, and bicubic is the sharpest. At augmentation scale, with small warps of 32×32 images, the difference is much smaller. We test whether it changes accuracy in the ablation later. Now Abdul Rafeh will explain our method.

ROMAN URDU

Same warp, teen interpolators: 24×24 crop, ~10 guna zoom, 10 degree rotation. Nearest blocky, bilinear smooth, bicubic sab se sharp. Augmentation mein chhote warps aur 32×32 images par fark kahin kam hota hai; ablation mein dekhenge ke accuracy par asar parta hai ya nahi. Ab Abdul Rafeh hamara method samjhayenge.

Slide 13 · Section: Methodology

Abdul Rafeh

ENGLISH

Thank you. Next, how we put geometric and photometric augmentation inside a CNN training pipeline.

ROMAN URDU

Shukriya. Ab ye ke humne geometric aur photometric augmentation ko CNN ki training pipeline mein kaise lagaya.

Slide 14 · Geometric augmentation pipeline

Abdul Rafeh

ENGLISH

For every image in every epoch we build one random 3×3 matrix: $H = C \cdot P \cdot T \cdot R \cdot Sh \cdot S \cdot F \cdot C^{-1}$. F is a horizontal flip with probability one half. S is a scale between 0.85 and 1.15, Sh a shear of up to 10 degrees, R a rotation of up to 15 degrees, T a translation of up to 4 pixels, and P a small perspective term. Each one is a transformation from the lecture. C moves the origin to the image centre: to rotate about the centre we translate, rotate and translate back. We multiply everything into one homography and apply it by inverse warping with grid_sample interpolation, on the GPU.

ROMAN URDU

Har epoch mein har image ke liye ek random 3×3 matrix: $H = C \cdot P \cdot T \cdot R \cdot Sh \cdot S \cdot F \cdot C^{-1}$. F: horizontal flip, probability aadha. S: scale 0.85 se 1.15. Sh: shear 10 degree tak. R: rotation 15 degree tak. T: translation 4 pixel tak. P: chhota perspective. Har ek lecture ki transformation hai. C origin ko image ke center par le jata hai: center ke gird ghumane ke liye translate, rotate, wapis translate. Sab ko multiply kar ke ek homography, phir GPU par inverse warping aur grid_sample interpolation.

Slide 15 · Photometric augmentation

Abdul Rafeh

ENGLISH

Photometric augmentation changes pixel values, not positions. Brightness multiplies the image by beta. Contrast scales the image around its mean gray. Saturation scales it around its grayscale version. We add sensor noise with sigma up to 0.04, and 10% of images are made grayscale. Geometric augmentation answers: where is the object, how big is it, and from which angle? Photometric answers: what lighting, which camera, how much noise? Neither changes the label.

ROMAN URDU

Photometric pixel ki value badalta hai, position nahi. Brightness: image ko beta se multiply. Contrast: mean gray ke gird scale. Saturation: grayscale version ke gird scale. Sensor noise sigma 0.04 tak, aur 10% images grayscale. Geometric jawab deta hai: object kahan hai, kitna bara, kis angle se? Photometric: kaisi roshni, kaun sa camera, kitna noise? Dono label nahi badalte.

Slide 16 · What the CNN sees

Abdul Rafeh

ENGLISH

This is the real output of our GPU pipeline on CIFAR-10. Every epoch draws new random parameters, so 60 epochs give 60 different versions of each training image. The black corners come from the inverse warp falling outside the source, exactly like the rotated cat in the lecture. The label never changes: a tilted, darker plane is still a plane.

ROMAN URDU

Ye CIFAR-10 par hamari GPU pipeline ka asli output hai. Har epoch naye random parameters, to 60 epochs = har image ke 60 alag versions. Black kone inverse warp ke bahar girne se hain, bilkul lecture ki ghoomi hui billi jaise. Label kabhi nahi badalta: tircha aur andhera jahaz phir bhi jahaz hai.

Slide 17 · Experimental setup

Abdul Rafeh

ENGLISH

This is a controlled comparison: only the augmentation changes. Data: CIFAR-10, with 10,000 training images (a low-data regime), 5,000 held-out validation images, and the official 10,000 test images, never used in training. Model: a ResNet-9 style CNN with 6.5 million parameters and no dropout, so overfitting is visible. We use SGD with Nesterov momentum, a one-cycle learning rate up to 0.05, weight decay, batch size 256 and 60 epochs, on an RTX 4090. We compare four settings (none, geometric, photometric, both), each with 3 seeds, and test on 13 shifts. Training accuracy is measured on clean images, so the train–test gap is an honest measure of overfitting.

ROMAN URDU

Controlled comparison: sirf augmentation badalti hai. Data CIFAR-10: 10,000 training images (kam data), 5,000 validation alag, aur official 10,000 test images jo training mein kabhi use nahi hui. Model: ResNet-9 jaisa CNN, 6.5 million parameters, dropout nahi, taake overfitting dikhe. SGD Nesterov, one-cycle learning rate 0.05 tak, weight decay, batch 256, 60 epochs, RTX 4090 par. Chaar settings: none, geometric, photometric, dono; har ek 3 seeds; aur 13 shifts par test. Training accuracy clean images par naapi, isliye train–test gap overfitting ka imaandar measure hai.

Slide 18 · Section: Results

Abdul Rafeh

ENGLISH

Now the results: overfitting, accuracy and robustness on unseen data.

ROMAN URDU

Ab results: overfitting, accuracy, aur unseen data par robustness.

Slide 19 · Training curves

Abdul Rafeh

ENGLISH

These curves are the mean of 3 seeds. Without augmentation the network reaches 100% training accuracy while validation stalls near 79%: memorisation. The validation loss even starts rising, which is classic overfitting. With geometric plus photometric augmentation, training accuracy stays lower, at 98.6%, but validation rises to 86%, and the final validation loss falls from 0.65 to 0.43. Lower training accuracy here is a good sign: the model is not memorising.

ROMAN URDU

Ye 3 seeds ka average hai. Augmentation ke baghair training accuracy 100% ho jati hai lekin validation 79% ke qareeb ruk jati hai: ratta. Validation loss barhne lagta hai: classic overfitting. Geometric + photometric ke saath training accuracy kam rehti hai (98.6%) lekin validation 86% tak jati hai aur validation loss 0.65 se 0.43. Yahan kam training accuracy achhi nishani hai: model ratta nahi laga raha.

Slide 20 · Accuracy table

Abdul Rafeh

ENGLISH

On the unseen test set: with no augmentation, 78.5%; geometric, 83.6%; photometric, 77.9%; both, 85.4%. That is +6.9 points of test accuracy, the train–test gap shrinks from 21.5 to 13.2 points, and validation loss falls by 34%. The standard deviations are only 0.1 to 0.3, so this is not luck. Geometric augmentation gives most of the gain on clean CIFAR-10. Photometric alone does not help clean accuracy, because the test lighting matches training, but it adds robustness, as the next slide shows.

ROMAN URDU

Unseen test set par: augmentation ke baghair 78.5%, geometric 83.6%, photometric 77.9%, dono 85.4%. Yani test accuracy +6.9 points, train–test gap 21.5 se 13.2, validation loss 34% kam. Standard deviation sirf 0.1–0.3, to ye luck nahi. Clean CIFAR-10 par zyada faida geometric se. Photometric akela clean accuracy nahi barhata kyunki test ki lighting training jaisi hai, lekin robustness deta hai: agli slide.

Slide 21 · Robustness under 13 shifts

Abdul Rafeh

ENGLISH

Here we apply 13 shifts to the test images and evaluate without retraining. The mean accuracy rises from 58.8% to 81.3%. On rotated and dark images it goes from 38.7% to 80.8%, a gain of 42 points. Most importantly, blur was never used in training, yet accuracy on blurred images rises from 43.4% to 72.7%. So the model learned more general features, not just the specific shifts it was trained on.

ROMAN URDU

Test images par 13 shifts laga kar baghair retraining evaluate kiya. Average accuracy 58.8% se 81.3%. Ghoomi + andheri images par 38.7% se 80.8%: 42 points. Sab se important: blur training mein kabhi use nahi hua, phir bhi blurry images par 43.4% se 72.7%. Matlab model ne zyada general features seekhe, sirf wo shifts nahi jin par train hua.

Slide 22 · Which augmentation buys which robustness

Abdul Rafeh

ENGLISH

Each augmentation buys the robustness it simulates. Geometric augmentation lifts the geometric shifts (rotation, zoom, shear, perspective) from 64.3% to 82.1%. Photometric lifts the lighting and noise shifts from 55.9% to 73.0%. Combining both covers both, and also the combined rotate-and-dark shift: 38.7% to 80.8%.

ROMAN URDU

Har augmentation wohi robustness deti hai jo wo simulate karti hai. Geometric: rotation, zoom, shear, perspective wale shifts 64.3% se 82.1%. Photometric: roshni aur noise wale shifts 55.9% se 73.0%. Dono milein to dono milte hain, aur combined rotate + dark bhi 38.7% se 80.8%.

Slide 23 · Ablations

Abdul Rafeh

ENGLISH

Two ablations. First, the interpolator inside the warp: nearest 85.9%, bilinear 85.3%, bicubic 86.4%. Bicubic is best, but the spread is only 1.1 points, from a single seed, and far smaller than the 6.9-point gain from augmenting at all. Second, data size: augmentation adds 6.9 points with 10,000 images and still 4.7 points with 45,000. Augmented 10k, at 85.4%, recovers much of what 4.5 times more real data gives, which is 87.7%. Now Mahmoob will compare our method with the state of the art.

ROMAN URDU

Do ablations. Pehla, warp ke andar interpolator: nearest 85.9, bilinear 85.3, bicubic 86.4. Bicubic best lekin fark sirf 1.1 points, ek seed, aur augmentation lagane ke 6.9 points se kahin kam. Doosra, data size: 10,000 images par +6.9, 45,000 par bhi +4.7. Augmented 10k (85.4%) taqreeban utna jitna 4.5 guna zyada asli data (87.7%). Ab Mahmoob state of the art se muqabla dikhayenge.

Slide 24 · Section: State-of-the-art benchmark

Mahmoob

ENGLISH

Thank you. A fair question is whether our pipeline is competitive with the augmentation methods used in current research, so we benchmarked it.

ROMAN URDU

Shukriya. Jaiz sawal hai ke kya hamari pipeline aaj ki research wale methods ke muqable ki hai, isliye humne benchmark kiya.

Slide 25 · Modern augmentation methods

Mahmoob

ENGLISH

We compare 9 methods. The standard baseline is random crop plus flip. AutoAugment learns a policy with reinforcement learning. RandAugment removes the search: 2 random operations at one magnitude. TrivialAugment applies one random operation at a random strength, with no parameters at all. AugMix mixes augmentation chains and adds a Jensen–Shannon consistency loss for robustness. Mixup blends two images, and CutMix pastes a patch, with the labels mixed in proportion. Finally, a combined modern recipe. Every method shares one model, optimiser, schedule and data split; only the augmentation changes. The policy methods use torchvision's reference implementations.

ROMAN URDU

9 methods. Standard baseline: random crop + flip. AutoAugment: reinforcement learning se policy seekhta hai. RandAugment: search khatam, 2 random operations ek magnitude par. TrivialAugment: ek random operation random strength par, koi parameter nahi. AugMix: augmentation chains mix karta hai aur robustness ke liye Jensen–Shannon consistency loss. Mixup: do images blend; CutMix: patch paste; labels ratio se mix. Aur ek combined modern recipe. Sab ka model, optimizer, schedule aur data same; sirf augmentation alag. Policy methods torchvision ki reference implementations hain.

Slide 26 · Benchmark protocol

Mahmoob

ENGLISH

We used the full CIFAR-10 training set: 45,000 train, 5,000 validation and 10,000 test. For robustness we used CIFAR-10-C, the standard benchmark from Hendrycks and Dietterich, ICLR 2019: 15 corruptions at 5 severities, which is 750,000 test images. Each method was trained with 2 seeds. We report clean accuracy, accuracy with flip test-time augmentation, calibration error (ECE), and mean corruption accuracy on CIFAR-10-C.

ROMAN URDU

Poora CIFAR-10: 45,000 train, 5,000 validation, 10,000 test. Robustness ke liye CIFAR-10-C, Hendrycks aur Dietterich (ICLR 2019) ka standard benchmark: 15 corruptions, 5 severities, 750,000 test images. Har method 2 seeds. Report: clean accuracy, flip test-time augmentation ke saath accuracy, calibration error (ECE), aur CIFAR-10-C par mean corruption accuracy.

Slide 27 · Results table

Mahmoob

ENGLISH

On clean accuracy, AutoAugment and RandAugment lead at about 94.7%; ours reaches 93.2%. On CIFAR-10-C, AugMix is best at 87.0% and ours is second at 86.0%, ahead of AutoAugment, RandAugment and TrivialAugment, and 13 points above the crop-and-flip baseline. On our own 13 shifts ours is first, but that test favours us, because it uses the same kinds of transforms we train on. Mixup and CutMix have high calibration error because they train on mixed labels and become under-confident, which is also reported in the literature.

ROMAN URDU

Clean accuracy mein AutoAugment aur RandAugment aage, ~94.7%; hamara 93.2%. CIFAR-10-C par AugMix best 87.0%, hamara doosra number 86.0%: AutoAugment, RandAugment aur TrivialAugment se aage, aur crop + flip baseline se 13 points zyada. Hamare apne 13 shifts par hum pehle hain, lekin wo test hamare haq mein hai kyunki wahi transforms hain jin par train kiya. Mixup aur CutMix ka calibration error zyada hai kyunki mixed labels par train ho kar model kam confident ho jata hai; literature mein bhi yehi reported hai.

Slide 28 · Clean vs corrupted accuracy

Mahmoob

ENGLISH

This chart compares clean and corrupted accuracy for every method. Every method loses accuracy under corruption, by 7 to 24 points. Our pipeline loses only about 7 points, the smallest drop, and ranks second of nine on CIFAR-10-C, at 86.0%.

ROMAN URDU

Ye chart har method ki clean aur corrupted accuracy dikhata hai. Har method corruption par accuracy khota hai, 7 se 24 points tak. Hamari pipeline sirf ~7 points khoti hai, sab se kam, aur CIFAR-10-C par 9 mein se doosre number par hai: 86.0%.

Slide 29 · By corruption family and severity

Mahmoob

ENGLISH

Where does the robustness come from? Our largest gain over the baseline is on noise, +28 points. To be fair, our photometric pipeline adds Gaussian noise, so that family is partly seen in training. If we remove the noise family, we are still in the top three, level with TrivialAugment. AugMix is strongest on blur and digital corruptions. At the highest severity, AugMix keeps 79% and ours 77.5%, against 57% for the baseline. Our weak spot is the contrast corruption, because our contrast range is mild.

ROMAN URDU

Robustness kahan se aati hai? Baseline par hamara sab se bara faida noise par: +28 points. Imaandari se: hamari photometric pipeline Gaussian noise lagati hai, to wo family kuch had tak training mein dekhi hui hai. Noise family nikal dein to bhi hum top-3 mein hain, TrivialAugment ke barabar. AugMix blur aur digital par sab se strong. Sab se zyada severity par AugMix 79%, hum 77.5%, baseline 57%. Hamari kamzori contrast corruption hai, kyunki hamari contrast range halki hai.

Slide 30 · What the benchmark shows

Mahmoob

ENGLISH

In summary: AutoAugment gives the best clean accuracy (94.7%), AugMix the best robustness (87.0%) and the best calibration. Five of eight methods beat crop and flip on clean accuracy. Our from-scratch pipeline ranks 8th on clean accuracy, 2nd on CIFAR-10-C and 1st on our shifts. It is a trade-off: about 1.5 points of clean accuracy for a large gain in robustness. It is also cheap: about 4 minutes per run, against 15 for AugMix.

ROMAN URDU

Khulasa: AutoAugment ki clean accuracy best (94.7%), AugMix ki robustness (87.0%) aur calibration best. Aath mein se paanch methods clean accuracy mein crop + flip se aage. Hamari scratch se bani pipeline clean mein 8th, CIFAR-10-C par 2nd, hamare shifts par 1st. Ye trade-off hai: ~1.5 points clean accuracy ke badle bohat zyada robustness. Aur sasti: ~4 minute per run, AugMix 15 minute.

Slide 31 · Key takeaways

Mahmoob

ENGLISH

Three takeaways. One: the lecture's matrices are the augmentation, from translation to homography, applied by inverse warping with interpolation. Two: augmentation reduces overfitting, adding 6.9 points of test accuracy and shrinking the train–test gap by 8.3 points. Three: augmentation makes inference robust, by 22.5 points under our shifts, including blur, which was never trained on. Against the state of the art, our pipeline is second of nine on CIFAR-10-C.

ROMAN URDU

Teen baatein. Ek: lecture ki matrices hi augmentation hain: translation se homography tak, inverse warping aur interpolation se. Do: augmentation overfitting kam karti hai: test accuracy +6.9 points, gap 8.3 points kam. Teen: inference robust hoti hai: hamare shifts par +22.5 points, blur samait jo kabhi train nahi hua. State of the art ke muqable mein CIFAR-10-C par 9 mein se doosre number par.

Slide 32 · Limitations & future work

Mahmoob

ENGLISH

Limitations: one dataset, one architecture and a short 60-epoch schedule. Papers train wider networks for 200 epochs. Our own shifts overlap with our augmentations, which is why CIFAR-10-C is the fairer test. Large rotations or flips can change the label for digits or text, like 6 and 9. Future work: bigger models and ImageNet-C, learnable warp ranges, and test-time augmentation.

ROMAN URDU

Limitations: ek dataset, ek architecture, sirf 60 epochs; papers 200 epochs tak bare networks train karte hain. Hamare shifts hamari augmentations se overlap karte hain, isliye CIFAR-10-C fair test hai. Bari rotation ya flip digits/text ka label badal sakte hain, jaise 6 aur 9. Future: bare models aur ImageNet-C, seekhne wali warp ranges, aur test-time augmentation.

Slide 33 · Code walkthrough

Mahmoob

ENGLISH

This is inverse warping in eight lines. For every output pixel we build homogeneous coordinates, multiply by H inverse to find where it came from, divide by w, normalise to the −1 to 1 range, and let grid_sample interpolate. To reproduce everything: demo_transforms.py makes all the theory figures in seconds, and run_all.sh runs the full experiment.

ROMAN URDU

Aath lines mein inverse warping. Har output pixel ke homogeneous coordinates, H inverse se multiply kar ke source location, w se divide, −1 se 1 mein normalize, aur grid_sample interpolate karta hai. Sab dobara chalane ke liye: demo_transforms.py seconds mein saari theory figures, aur run_all.sh poora experiment.

Slide 34 · References

Mahmoob

ENGLISH

Our main references are Sir's Lecture 7, Szeliski, and Hartley and Zisserman for the DLT. Keys covers bicubic interpolation, and the original papers cover each augmentation method and CIFAR-10-C.

ROMAN URDU

Hamare references: Sir ka Lecture 7, Szeliski, aur DLT ke liye Hartley–Zisserman; bicubic ke liye Keys; aur har augmentation method aur CIFAR-10-C ke asli papers.

Slide 35 · Questions

All

ENGLISH

Thank you. We are happy to take your questions.

ROMAN URDU

Shukriya. Aap ke sawalat ka intezar hai.

Part B · Lecture 7 transcript (where the topic was assigned)

Original lecture in Roman Urdu / English, lightly cleaned (repeated words removed), with a full English translation of each part.

1. Recap: interpolation and warping

Roman Urdu (original)

Okay, so last time humne discuss kiya tha interpolation, aur maine aapko uska background bataya tha ke why we need interpolation. Because we do image transformations, and at the destination pixel you need the intensity values. Toh usme humne 2 methods study kiye the: ek tha forward warping, aur doosra tha inverse warping. Toh inverse warping mein, hum source grid ke jo pixels hain unko use karte hain as neighbours to interpolate some value. Bilinear interpolation maine aapko thoda detail se samjhaya tha. Bicubic jo hai wo polynomial-based hai; usme calculations thodi complex ho jati hain, so we don't need to go into those details.

English

Last time we discussed interpolation, and I explained why we need it: when we transform an image, we need intensity values at the destination pixels. We studied two methods, forward warping and inverse warping. In inverse warping, we use the pixels of the source grid as neighbours to interpolate a value. I explained bilinear interpolation in some detail. Bicubic is polynomial-based; its calculations are more complex, so we don't need to go into those details.

2. Interpolation results

Roman Urdu (original)

So any questions up till now? These were the results... but these results really depend upon the images you are choosing. This is one of the examples, not a great example. But still, there are some small lines, thik hai? Jis ko aap dekh sakte hain ke how it is affecting image interpolation.

English

Any questions so far? These were the results, but they really depend on which images you choose. This is one example, not a great one, but there are some small lines in it where you can see how the choice of interpolation affects the image.

3. Rigid transformation and degrees of freedom

Roman Urdu (original)

Next, aap combine kar sakte hain different types of transformations, jaise ye rigid body transformation hai. Isme agar aap dekhein, toh you have rotation and translation combined. Whatever transformation you are doing, you need to do inverse warping to find the values at the destination pixels. Now you can also imagine ke transformation matrix ke kitne degrees of freedom hain, so how many points you need to solve this matrix? You have 3 degrees of freedom... so you need 2 points; translation ke liye aapko 1 point chahiye. Toh point nahi hoga, wo actually **point pair** hoga. Isi tarah aap ne sochna hai ke aap isko kaise solve karte hain. You can do other kinds of transformations, like here you just add scale also: scale is another parameter.

English

Next, you can combine different types of transformations, such as this rigid-body transformation: rotation and translation combined. Whatever transformation you apply, you need inverse warping to find the values at the destination pixels. Now think about how many degrees of freedom the matrix has, and how many points you need to solve it. There are 3 degrees of freedom, so you need 2 points; for translation alone you need 1. And it is not a point, it is actually a **point pair**. This is how you should think about solving it. You can do other transformations too, for example adding scale, which is one more parameter.

4. The topic: data augmentation on a CNN

Roman Urdu (original)

Toh jo image hai na, uski na ye saari augmentation hai. Toh ek presentation aap isme se bhi le sakte hain: Data Augmentation. But you have to go into the details, and also try to do it on some CNN. Jab aap CNN ki accuracy check karte hain kisi bhi dataset pe, toh aap ye kehte hain ke jo augmentation hai, that is going to make your inference more robust and more accurate. Aap 2-2 members ka group hain class mein. If they want to take this presentation on augmentation, they have to present it and show that it really works for CNNs. So please let me know if you want to do this.

English

All of these are augmentations of the image. So you can take a presentation from this too: Data Augmentation. But you have to go into the details, and also try it on a CNN. When you check a CNN's accuracy on any dataset, you claim that augmentation will make your inference more robust and more accurate. You are in groups of two. If a group wants to take this presentation on augmentation, they have to present it and show that it really works for CNNs. Please let me know if you want to do this.

5. Affine: six unknowns

Roman Urdu (original)

Agar aap isko thoda generic kar dein, kyunki abhi tak hum rotation, translation mein dekh rahe the, let's keep it for general purpose. Toh aap isme degrees of freedom barha sakte hain, jaise isme a, b, c, d, e, f parameters hain. Now you have 6 unknowns, and you need at least 3 corresponding point pairs to solve for 6 unknowns. Transformation is par depend karti hai ke a, b, c, d, e, f mein kya aata hai; wo transformation aapko image par milegi.

English

Let's make it more general, since so far we have looked only at rotation and translation. We can increase the degrees of freedom, for example with parameters a, b, c, d, e, f. Now there are 6 unknowns, and you need at least 3 corresponding point pairs to solve for them. The transformation you see on the image depends on the values of a to f.

6. Homography: eight unknowns, defined up to scale

Roman Urdu (original)

Coming back to what we are looking for: homography. Homography bhi 3×3 matrix se transformation hai, lekin isme degrees of freedom barha di: unknowns a, b, c, d, e, f, g, h. Jo last wala 1 hai, that is defining it up to a scale, jaise hum M matrix mein karte the, last element ko 1 kar dete the, kyunki projection onto the image is only correct up to a scale. Ab baaki saare unknowns use karein, toh image is tarah ke grid mein transform hogi. Ye strange-looking form hai, lekin jab pixels ko values milengi wo isi angle par milengi. Baaki image toh seedha square hota hai, side se black hoga. Is it a rectangle? No, parallelogram kind of a thing.

English

Coming back to what we are looking for: the homography. It is also a 3×3 transformation, but with more degrees of freedom: the unknowns a to h. The last entry, 1, defines it up to scale, just as we set the last element of the M matrix to 1, because the projection onto the image is only correct up to scale. Using all the other unknowns, the image transforms into a grid like this. It looks strange, but that is the angle at which the pixels get their values. The original image is a square, and the sides will be black. Is it a rectangle? No, more like a parallelogram.

7. Projective geometry: same centre of projection

Roman Urdu (original)

Toh it really depends upon your unknowns, but this is full flexibility: homography. Homography defines the projective transformation from one plane to the other. Ye projective geometry hai. Hum jaante hain ke hum camera ke center se pictures lete hain: camera ki location fixed hai. Ek location se picture lete hain, phir camera ka orientation halka sa change kar ke usi scene ki picture lete hain. Dono pictures mein kuch overlap hoga, kuch pixels ka overlap hoga, kuch ka nahi, lekin do planes milenge. Ye dono pictures same center of camera se li gayi hain. Toh is case mein ek projective plane ya ek image ko doosri image par kaise map karein, jo same camera, same center of projection se li gayi ho, but a little bit different...

English

So it really depends on your unknowns, but this is full flexibility: the homography. A homography defines the projective transformation from one plane to another; this is projective geometry. We know we take pictures from the camera's centre, with the camera location fixed. We take one picture, then change the camera's orientation slightly and take another of the same scene. The two pictures overlap: some pixels overlap and some don't, but we get two planes. Both pictures are taken from the same camera centre. So how can we map one projective plane, one image, to the other image taken from the same camera and centre of projection, but slightly different...

Task specification (as given)

Topic: Impact of Data Augmentation Techniques on CNN Performance and Inference Robustness.
Group constraint: maximum 2 members per group (*hamara group 3 ka hai: Sir se confirm kar lein*).
Scope: (1) Theoretical foundation: translation, rotation, scaling, affine, homography → augmentation. (2) Mathematical mechanics: forward vs inverse warping; nearest, bilinear, bicubic interpolation. (3) Implement geometric and photometric augmentation on a CNN pipeline. (4) Quantitative proof (training curves, accuracy, validation loss) that augmentation prevents overfitting and makes inference more accurate and robust on unseen test data.

Part C · 10 questions Sir is most likely to ask

The full bank of 50 questions is in the Roman Urdu project-explanation PDF.

1. How many point pairs for a rigid transform, and why?

EN: Two. It has 3 DoF and each pair gives 2 equations; one pair gives only 2, so we need 2 pairs.

UR: Do. 3 DoF hain, har pair 2 equations deta hai; ek pair kaafi nahi, isliye 2.

2. Why 8 DoF for a homography, not 9?

EN: It is defined up to scale: H and kH give the same mapping after dividing by w, so we fix $h_{33} = 1$.

UR: Scale tak defined hai: H aur kH same mapping; isliye $h_{33} = 1$ fix.

3. Why inverse warping instead of forward?

EN: Forward warping leaves holes (45.1% in our demo); inverse warping visits every target pixel once, with no holes.

UR: Forward mein holes (hamare demo mein 45.1%); inverse har target pixel ek baar visit karta hai, koi hole nahi.

4. Derive bilinear interpolation.

EN: Interpolate in x: R1 = (1−dx)Q11 + dx·Q21 and R2 = (1−dx)Q12 + dx·Q22. Then in y: f = (1−dy)R1 + dy·R2.

UR: Pehle x mein R1, R2; phir y mein f = (1−dy)R1 + dy·R2.

5. How is a homography solved from 4 pairs?

EN: By the DLT: 8 linear equations, A·h = 0. Take the SVD; h is the right singular vector with the smallest singular value, then divide by h_{33}.

UR: DLT: A·h = 0, SVD, sab se chhoti singular value wala vector, phir h_{33} se divide.

6. How do you prove reduced overfitting?

EN: The train–test gap falls from 21.5 to 13.2 points, validation loss falls by 34%, and test accuracy rises by 6.9 points. This holds over 3 seeds, and the test set was never used in training.

UR: Gap 21.5 se 13.2, validation loss 34% kam, test +6.9; 3 seeds; test set training mein kabhi nahi.

7. Isn't testing on shifts you trained on cheating?

EN: Partly, which is why we also report blur (never trained on: +29 points) and the independent CIFAR-10-C benchmark, where we rank 2nd of 9.

UR: Kuch had tak; isliye blur (kabhi train nahi: +29) aur CIFAR-10-C (2nd of 9) bhi report kiye.

8. Does the interpolation method matter for the CNN?

EN: Only a little: 85.3% to 86.4%, a 1.1-point spread from one seed, which is much smaller than the 6.9-point gain from augmentation itself.

UR: Thora sa: 85.3–86.4, 1.1 points; augmentation ke 6.9 points se kahin kam.

9. Is your project state of the art?

EN: We benchmarked against 8 modern methods. We are 2nd of 9 on CIFAR-10-C robustness (86.0% against AugMix's 87.0%) but about 1.5 points lower on clean accuracy. Part of our noise robustness comes from training with noise.

UR: 8 modern methods se compare kiya: robustness mein 2nd (86.0 vs AugMix 87.0), clean accuracy ~1.5 kam; noise par faida kuch had tak noise training se.

10. When can augmentation hurt?

EN: When it changes the label (6 ↔ 9, flipped text) or is so strong that the object is no longer recognisable.

UR: Jab label badal de (6 ↔ 9, ulta text) ya itni strong ho ke object pehchan mein na aaye.